Quantitative Macro Theory **HW #2**
Spring 2018 **Due at 3pm on Wed 1/31/2018**

Write your answer on latex and prepare a pdf file. The file should be clearly written and organized. You should think of that file as a short version of a paper. The appendix of that file should include your code. Email me that pdf file before the deadline.

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1. (100 points) Consider this problem:

$$V(k_i, \varepsilon_r) = \max_{c,k'} \frac{1}{1-\gamma}\left(c - \frac{\hat{l}^{1+\theta}}{1+\theta}\right)^{1-\gamma} + \beta \sum_{s=1}^{2} \pi_{rs} V(k'_j, \varepsilon_s),$$

subject to,

$$c = A(k_i \hat{h})^{\alpha} \hat{l}^{1-\alpha} - k'_j e^{-\varepsilon_r} + k_i \left(1 - \hat{h}^{\omega}/\omega\right) e^{-\varepsilon_r},$$

where

$$\hat{h}, \hat{l} = \arg\max \left[A(k_i \hat{h})^{\alpha} \hat{l}^{1-\alpha} - k_i \left(1 - \hat{h}^{\omega}/\omega\right) e^{-\varepsilon_r} - \frac{\hat{l}^{1+\theta}}{1+\theta} \right]$$

Calibration

Set $\beta = 0.96$ as in Kydland and Prescott (1982), $\alpha = 0.29$ to match the capital's share of income (NIPA), $1/\theta = 1.7$ to match the intertemporal elasticity of labor supply estimated by MaCurdy (1981) and others, $\gamma = 2$, and $\omega = 1.42$ to imply a rate of depreciation of 10% in equilibrium.

In terms of the stochastic process, set

$$\pi_{11} = \pi_{22} = \pi \text{ and } \varepsilon_1 = -\varepsilon_2 = \Theta.$$

Note that the standard deviation $\sigma = \Theta$ and the coefficient of autocorrelation is $\lambda = 2\pi - 1$. These parameters are picked so as to make the model generate the same standard deviation and first-order serial correlation for output as is observed in the data. $\sigma = 0.051$ and $\lambda = 0.44$.

(Note: I think I covered all the parameters but if you think I am missing one, pick the value of your choice and let me know.)

(a) (20 points) Solve the model using value function iteration + discretization. Plot V and k' as a function of k and include two lines, one for each value of the shock ε.

(b) (10 points) Implement the tricks that we talked about in class that take advantage of the fact that the policy function is increasing and the maximand is concave. How much time did you save with each of them?

(c) (10 points) Implement the Howard iteration approach to save time. Report how much time did you save and what was the optimal value of n_h that you found.

Table 1: Key data moments

Variable	Standard deviation	Correlation with output	Autocorrelation
Output			
Consumption			
Investment			
Hours worked			
Productivity			
Utilization			

(d) (20 points) Look for data for time series of output, consumption, investment hours worked, productivity, and capital utilization for the US for the postwar period. Do a table like Table 1 and report the sources very carefully. Before computing the statistics detrend the series using the HP method or other.

(e) (20 points) Compute these statistics using the model. Compare them with the data. Summarize your findings.

(f) (20 points) Find data for any developing country of your choice and repeat Table 1. Recalibrate σ and λ to obtain a good fit of the autocorrelation and standard deviation of output for that country. Compute the statistics from the model again. Do you do a good job at accounting for fluctuations in that country?